

THREE HILLS, AB, Canada

WMO: 712490

Lat: 51.7675N

Lon: 113.2056W

Elev: 857

StdP: 91.44

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.9	(c) -26.7	(d) -33.0	(e) 0.2	(f) -29.8	(g) -29.6	(h) 0.3	(i) -26.5	(j) 13.8	(k) -5.9	(l) 12.1	(m) -5.8	(n) 3.2	(o) 270	(p) 0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	29.4	17.0	27.3	16.4	25.4	15.7	19.2	25.9	18.0	24.6	16.9	23.3	4.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	13.4	22.5	15.6	12.3	21.0	14.4	11.4	19.7	58.6	26.0	54.4	24.7	50.8	23.4	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	10.3	9.3	-33.4	32.6	3.5	2.2	-35.9	34.2	-38.0	35.5	-39.9	36.7	-42.5	38.3
			-33.5	21.8	3.5	1.6	-36.0	22.9	-38.1	23.9	-40.0	24.7	-42.5	25.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-10.2	-8.8	-3.6	3.9	10.1	14.1	16.8	15.9	11.4	4.4	-3.3	-9.3
	DBStd	11.52	8.79	7.83	7.97	5.31	4.55	3.12	3.02	3.40	4.59	5.11	7.19	8.08
	HDD10.0	3044	627	527	422	190	54	4	1	2	37	183	399	599
	HDD18.3	5451	885	760	679	432	256	132	67	88	212	432	648	858
	CDD10.0	675	0	0	0	58	126	210	186	78	9	1	0	0
	CDD18.3	41	0	0	0	1	4	18	14	3	0	0	0	0
	CDH23.3	923	0	0	0	3	64	88	320	327	117	4	0	0
	CDH26.7	239	0	0	0	0	10	13	91	97	28	0	0	0
(14) Wind	WSAvg	4.3	4.3	4.2	4.3	4.9	5.1	4.7	3.8	3.7	4.2	4.4	4.2	4.1
(15) Precipitation	PrecAvg	407	18	14	18	24	45	70	72	49	46	17	14	16
	PrecMax	577	57	32	49	60	111	170	154	125	126	46	46	41
	PrecMin	303	0	0	0	0	10	19	16	7	6	1	2	0
	PrecStd	71	16	9	12	15	26	36	37	32	30	12	11	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	8.0	15.4	22.8	28.1	28.5	31.7	32.3	30.0	23.3	15.1	9.5
		MCWB	3.5	3.7	7.3	10.4	13.8	16.3	18.4	17.2	14.8	11.7	7.7	3.7
	2%	DB	5.1	5.2	11.1	19.4	25.1	25.8	29.2	29.3	26.6	18.6	11.2	6.2
		MCWB	2.2	1.9	5.2	9.2	12.4	15.3	18.8	16.5	14.1	9.8	5.6	2.6
	5%	DB	3.7	3.3	8.4	16.3	22.3	23.7	26.8	27.0	23.9	15.6	8.3	3.9
		MCWB	1.3	0.8	3.9	7.5	11.5	14.5	18.1	16.4	13.2	8.3	3.8	1.0
	10%	DB	1.6	1.2	5.8	13.2	19.7	21.8	24.6	24.7	20.8	12.8	5.7	1.7
		MCWB	-0.3	-0.7	2.6	6.0	10.3	13.6	17.1	16.0	12.1	7.0	2.2	-0.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	4.0	8.1	11.5	14.8	18.6	21.8	20.1	16.2	12.5	8.2	4.6
		MCDB	6.7	7.3	14.2	21.0	23.3	25.4	26.6	26.4	26.2	21.3	14.5	8.4
	2%	WB	2.5	2.1	5.7	9.7	13.4	16.8	20.0	18.6	14.8	10.4	6.0	2.8
		MCDB	4.8	4.7	10.4	18.5	22.6	22.9	26.5	25.4	24.4	17.5	10.7	5.7
	5%	WB	1.4	0.8	4.1	8.0	12.1	15.6	18.8	17.5	13.6	8.9	4.2	1.1
		MCDB	3.6	3.1	8.0	15.0	20.7	21.7	25.4	24.5	22.4	14.7	7.9	3.8
	10%	WB	-0.3	-0.6	2.6	6.4	10.9	14.4	17.6	16.3	12.5	7.4	2.4	-0.5
		MCDB	1.6	1.2	5.6	12.3	18.6	20.4	23.7	23.3	20.3	12.3	5.2	1.7
(35) Mean Daily Temperature Range	MDBR		9.4	10.0	10.4	12.6	13.5	12.1	13.7	14.4	14.0	12.5	10.1	9.5
	5% DB	MCDBR	10.7	11.4	13.9	18.3	18.1	16.1	17.5	19.2	19.6	17.1	13.5	11.3
		MCWBR	8.3	8.8	8.9	9.7	8.3	7.5	9.0	8.7	8.9	9.4	8.7	8.3
	5% WB	MCDBR	10.5	11.1	13.4	16.7	16.2	14.1	16.3	16.5	17.8	16.3	13.1	11.0
		MCWBR	8.5	8.7	8.9	9.2	7.6	7.4	9.2	8.5	8.5	9.2	8.7	8.2
(40) Clear-Sky Solar Irradiance	taub		0.264	0.275	0.286	0.331	0.348	0.349	0.348	0.363	0.316	0.283	0.253	0.243
	taud		2.332	2.391	2.448	2.351	2.355	2.392	2.409	2.359	2.454	2.528	2.479	2.358
	Ebn at Noon		759	863	923	908	904	904	898	863	874	848	787	732
	Edh at Noon		61	77	88	110	116	113	109	109	88	66	52	50
(44) All-Sky Solar Radiation	RadAvg		1.09	2.12	3.51	4.74	5.74	6.13	6.51	5.28	3.71	2.23	1.18	0.84
	RadStd		0.09	0.17	0.29	0.39	0.37	0.37	0.33	0.37	0.39	0.26	0.10	0.08

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	+282	+313	N/A	-19
(47) Regional (73 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page